Roll No. | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 13

B.Pharma (Sem.–3)

PHYSICAL PHARMACEUTICS-I

Subject Code : BP-302T

M.Code : 75106

Date of Examination : 14-12-22

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Answer briefly :

a. Solutions

b. Raoult's Law

c. Amorphrous

d. Dipole moment

e. HLB

f. Surface tension

g. Buffers

h. Drug

i. Isotonic solutions

j. Complexation.

SECTION-B

2. a. Give methods for the determination of surface and interfacial tension.

   b. Explain methods for the determination of CMC.

3. Classify complexation. Highlight methods of its analysis.

4. What are the ideal solubility parameters? Explain mechanism of solute-solvent interaction.

SECTION-C

5. Explain the quantitative approach to the factors influencing solubility of drugs.

6. Write short note on eutectic mixture and sublimation.

7. How refractive index and optical rotation determined.

8. Explain adsorption at solid interface.

9. How pH determined? Give its significance.

10. What is Buffer Capacity? Explain with example.

11. Explain the crystalline structure of complexes.

12. Explain surface active agents with examples.

13. Write short note on :

    a) Inhalers

    b) Critical solution temperature.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.